



18.0

SE-Standard Edition

Statistics and Data Science

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Notes:

1. Unicode is supported; see [help unicode advice](#).

2. Maximum number of variables is set to 5,000 but can be increased; see [help set maxvar](#).

1 .

2 . import excel "/Users/vahid/Documents/ReCod.xlsx", sheet("ReCod") firstrow
(39 vars, 50,161 obs)

3 . *How does training improve individual forecasts? Modeling Differences in Compensatory and Non-Compensatory Biases in *Geopolitical Forecasts

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17 .

18 .

19 . *Total Timer on:

20 . timer on 9

21 . *Model 1 with timer:

22 . timer on 1

23 . set seed 6092024

24 . bayes: logit BinaryOutcome c.ForecastLogRatio if Experience == 0

Burn-in ...
Simulation ...

Model summary

Likelihood:
BinaryOutcome ~ logit(xb_BinaryOutcome)

Prior:
{BinaryOutc~e:ForecastLogRatio _cons} ~ normal(0,10000) (1)

(1) Parameters are elements of the linear form xb_BinaryOutcome.

Bayesian logistic regression
Random-walk Metropolis-Hastings sampling

MCMC iterations = 12,500
Burn-in = 2,500
MCMC sample size = 10,000
Number of obs = 39,481
Acceptance rate = .1657
Efficiency: min = .08137
avg = .08514
max = .08891

Log marginal-likelihood = -19772.418

Equal-tailed

```

    BinaryOutcome |      Mean   Std. dev.   MCSE   Median [95% cred. interval]
ForecastLogRatio
  _cons          .3428654   .0069432   .000243   .3428856   .3297828   .3571815
              -1.040546   .0128871   .000432  -1.039804  -1.066064  -1.016002

Note: Default priors are used for model parameters.
Note: Adaptation tolerance is not met.

25 . timer off 1

26 . timer list 1
    1:      20.51 /      1 =      20.5060

27 .
28 .
29 . *Model 2 with timer:

30 . timer on 2

31 . set seed 6092024

32 . bayes: melogit BinaryOutcome c.ForecastLogRatio || user_id: c.ForecastLogRatio

Burn-in 2500 aaaaaaaaa1000aaaaaaaa2000aaaaa done
Simulation 10000 .....1000.....2000.....3000.....4000.....5000.....6000.....7000.....8000.....9000.....10000 done

Multilevel structure
-----
user_id
{U0}: random intercepts
{U1}: random coefficients for ForecastLogRatio
-----

Model summary
-----
Likelihood:
  BinaryOutcome ~ melogit(xb_BinaryOutcome)

Priors:
  {BinaryOutcome:ForecastLogRatio _cons} ~ normal(0,10000)          (1)
  {U0} ~ normal(0,{U0:sigma2})          (1)
  {U1} ~ normal(0,{U1:sigma2})          (1)

Hyperpriors:
  {U0:sigma2} ~ igamma(.01,.01)
  {U1:sigma2} ~ igamma(.01,.01)
-----
(1) Parameters are elements of the linear form xb_BinaryOutcome.

Bayesian multilevel logistic regression          MCMC iterations =      12,500
Random-walk Metropolis-Hastings sampling          Burn-in          =      2,500
                                           MCMC sample size =     10,000
Group variable: user_id          Number of groups=      851
                                           Obs per group:
                                           min =      1
                                           avg =     58.9
                                           max =     181
Family: Bernoulli          Number of obs =     50,161
Link: logit          Acceptance rate =     .3158
                                           Efficiency: min =     .001603
                                           avg =     .01709
                                           max =     .0527
Log marginal-likelihood

-----
      Mean   Std. dev.   MCSE   Median   Equal-tailed
      [95% cred. interval]
BinaryOutcome
ForecastLogRatio
  _cons      .5603127   .0143024   .002132   .5610709   .5301141   .5878185
              -.9920449   .0117032   .00051   -.9918111  -1.017125  -.9699269

user_id
  U0:sigma2   .0035771   .0015821   .000395   .0032093   .0014115   .0069379
  U1:sigma2   .0833537   .0076594   .000784   .0831036   .0688809   .0997087

Note: Default priors are used for model parameters.
Note: There is a high autocorrelation after 500 lags.
Note: Adaptation tolerance is not met in at least one of the blocks.

33 . timer off 2

34 . timer list 2
    2:      275.37 /      1 =      275.3710

35 .
36 .
37 . *Model 3 with timer:

38 . timer on 3

39 . set seed 6092024

40 . bayes: melogit BinaryOutcome c.ForecastLogRatio##i.TrainingBinary || user_id: c.ForecastLogRatio
```

How does training improve individual forecasts? Modeling Differences in Compensatory and Non-Compensatory Biases in Geopolitical Forecas User: Vahid karimi Motahhar Project: TrainingForecasting

Burn-in 2500 aaaaaaaaa1000aaaaaaaa2000aaaaa done

Simulation 100001000.....2000.....3000.....4000.....5000.....6000.....7000.....8000.....9000.....10000 done

Multilevel structure

user_id

{U0}: random intercepts

{U1}: random coefficients for ForecastLogRatio

Model summary

Likelihood:

BinaryOutcome ~ melogit(xb_BinaryOutcome)

Priors:

{BinaryOutc~e:ForecastLogRatio} ~ normal(0,10000)

(1)

{BinaryOutc~e:1.TrainingBinary} ~ normal(0,10000)

(1)

{BinaryOutc~e:1.TrainingBinary#c.ForecastLogRatio} ~ normal(0,10000)

(1)

{BinaryOutc~e:_cons} ~ normal(0,10000)

(1)

{U0} ~ normal(0,{U0:sigma2})

(1)

{U1} ~ normal(0,{U1:sigma2})

(1)

Hyperpriors:

{U0:sigma2} ~ igamma(.01,.01)

{U1:sigma2} ~ igamma(.01,.01)

(1) Parameters are elements of the linear form xb_BinaryOutcome.

Bayesian multilevel logistic regression

Random-walk Metropolis-Hastings sampling

Group variable: user_id

MCMC iterations = 12,500

Burn-in = 2,500

MCMC sample size = 10,000

Number of groups= 851

Obs per group:

min = 1

avg = 58.9

max = 181

Family: Bernoulli

Link: logit

Number of obs = 50,161

Acceptance rate = .2878

Efficiency: min = .001755

avg = .01554

max = .04418

Log marginal-likelihood

		Mean	Std. dev.	MCSE	Median	Equal-tailed [95% cred. interval]	
BinaryOutcome	ForecastLogRatio	.4900006	.0207033	.003214	.4884102	.4496442	.5338705
	1.TrainingBinary	.0159932	.0238054	.001133	.0158546	-.0312655	.0605214
TrainingBinary#c.ForecastLogRatio	1	.145275	.0283638	.004093	.1452252	.0874597	.2027102
	_cons	-.9991322	.0155361	.000995	-.9991846	-1.029551	-.9687961
user_id	U0:sigma2	.003008	.0011281	.000269	.00278	.0014401	.0058978
	U1:sigma2	.0785223	.0074615	.000631	.0777832	.0655406	.095103

Note: Default priors are used for model parameters.

Note: There is a [high autocorrelation](#) after 500 lags.

41 . timer off 3

42 . timer list 3

3: 406.14 / 1 = 406.1360

43 .

44 .

45 . *Model 4 with timer:

46 . timer on 4

47 . set seed 6092024

48 . bayes: melogit BinaryOutcome c.ForecastLogRatio##1.TrainingBinary c.ForecastLogRatio##1.year2binary || user_id: c.ForecastLogRatio

Burn-in 2500 aaaaaaaaa1000aaaaaaaa2000aaaaa done

Simulation 100001000.....2000.....3000.....4000.....5000.....6000.....7000.....8000.....9000.....10000 done

Multilevel structure

user_id

{U0}: random intercepts

{U1}: random coefficients for ForecastLogRatio

Model summary

Likelihood:

BinaryOutcome ~ melogit(xb_BinaryOutcome)

```
Priors:
      {BinaryOutc~e:ForecastLogRatio} ~ normal(0,10000)          (1)
      {BinaryOutc~e:1.TrainingBinary} ~ normal(0,10000)         (1)
{BinaryOutc~e:1.TrainingBinary#c.ForecastLogRatio} ~ normal(0,10000) (1)
      {BinaryOutc~e:1.year2binary} ~ normal(0,10000)            (1)
{BinaryOutc~e:1.year2binary#c.ForecastLogRatio} ~ normal(0,10000) (1)
      {BinaryOutc~e:_cons} ~ normal(0,10000)                    (1)
      {U0} ~ normal(0,{U0:sigma2})                              (1)
      {U1} ~ normal(0,{U1:sigma2})                              (1)
```

```
Hyperpriors:
{U0:sigma2} ~ igamma(.01,.01)
{U1:sigma2} ~ igamma(.01,.01)
```

(1) Parameters are elements of the linear form xb_BinaryOutcome.

```
Bayesian multilevel logistic regression          MCMC iterations =    12,500
Random-walk Metropolis-Hastings sampling         Burn-in       =     2,500
                                                MCMC sample size =   10,000
Group variable: user_id                        Number of groups=     851
                                                Obs per group:
min =          1
avg =         58.9
max =         181
Family: Bernoulli                             Number of obs   =   50,161
Link:    logit                                Acceptance rate =    .3059
                                                Efficiency: min =   .002575
                                                avg =         .01002
                                                max =         .02152
```

Log marginal-likelihood

		Mean	Std. dev.	MCSE	Median	Equal-tailed [95% cred. interval]	
BinaryOutcome	ForecastLogRatio	.5361353	.0193571	.003352	.5374144	.4933332	.5723646
	1.TrainingBinary	.0127702	.0232881	.00337	.0129111	-.032876	.0560048
TrainingBinary#c.ForecastLogRatio	1	.1509684	.0266868	.004813	.1512565	.1000039	.2012054
	1.year2binary	.1326383	.0233069	.001589	.1329748	.0869525	.1785681
year2binary#c.ForecastLogRatio	1	-.0710617	.0154281	.001232	-.0712995	-.1012413	-.040983
	_cons	-1.077495	.0207381	.001682	-1.077668	-1.117566	-1.035148
user_id	U0:sigma2	.0033198	.001035	.000204	.0032881	.001584	.0057165
	U1:sigma2	.0797102	.0072334	.000612	.0793784	.0667188	.095034

Note: Default priors are used for model parameters.
Note: There is a high autocorrelation after 500 lags.

49 . timer off 4

50 . timer list 4
4: 558.53 / 1 = 558.5340

51 .
52 .
53 . *Model 5 with timer:

54 . timer on 5

55 . set seed 6092024

56 . bayes: melogit BinaryOutcome c.ForecastLogRatio##i.TrainingBinary##1.year2binary if Experience == 0 || user_id: c.ForecastLogRatio

Burn-in 2500 aaaaaaaaa1000aaaaaaaa2000aaaaa done
Simulation 100001000.....2000.....3000.....4000.....5000.....6000.....7000.....8000.....9000.....10000 done

Multilevel structure

```
user_id
{U0}: random intercepts
{U1}: random coefficients for ForecastLogRatio
```

Model summary

```
Likelihood:
BinaryOutcome ~ melogit(xb_BinaryOutcome)
```

```
Priors:
      {BinaryOutc~e:ForecastLogRatio} ~ normal(0,10000)          (1)
      {BinaryOutc~e:1.TrainingBinary} ~ normal(0,10000)         (1)
{BinaryOutc~e:1.TrainingBinary#c.ForecastLogRatio} ~ normal(0,10000) (1)
      {BinaryOutc~e:1.year2binary} ~ normal(0,10000)            (1)
{BinaryOutc~e:1.year2binary#c.ForecastLogRatio} ~ normal(0,10000) (1)
      {BinaryOutc~e:1.TrainingBinary#1.year2binary} ~ normal(0,10000) (1)
```

```
{BinaryOutcome:1.TrainingBinary#1.year2binary#c.ForecastLogRatio} ~ normal(0,10000) (1)
{BinaryOutcome:_cons} ~ normal(0,10000) (1)
{U0} ~ normal(0,{U0:sigma2}) (1)
{U1} ~ normal(0,{U1:sigma2}) (1)
```

Hyperpriors:
{U0:sigma2} ~ igamma(.01,.01)
{U1:sigma2} ~ igamma(.01,.01)

(1) Parameters are elements of the linear form xb_BinaryOutcome.

Bayesian multilevel logistic regression	MCMC iterations =	12,500
Random-walk Metropolis-Hastings sampling	Burn-in =	2,500
	MCMC sample size =	10,000
Group variable: user_id	Number of groups =	851
	Obs per group:	
	min =	1
	avg =	46.4
	max =	111
Family: Bernoulli	Number of obs =	39,481
Link: logit	Acceptance rate =	.2977
	Efficiency: min =	.001895
	avg =	.007122
	max =	.01808

Log marginal-likelihood

		Mean	Std. dev.	MCSE	Median	Equal-tailed [95% cred. interval]	
BinaryOutcome	ForecastLogRatio	.5567402	.0257831	.004536	.5553023	.5071155	.613426
	1.TrainingBinary	-.0508664	.0261608	.00443	-.0504871	-.100387	-.0009611
	TrainingBinary#c.ForecastLogRatio						
	1	.1429439	.0351722	.005885	.1436615	.0667707	.2129311
	1.year2binary	.0742375	.0337885	.002801	.0731304	.0127713	.142892
	year2binary#c.ForecastLogRatio						
	1	-.0839671	.0298656	.004676	-.0851875	-.1350712	-.0257093
	TrainingBinary#year2binary						
TrainingBinary#year2binary#c.ForecastLogRatio	1 1	.1358649	.0465298	.004422	.1348107	.0441649	.2314608
	1 1	.0142111	.0216697	.004978	.0139568	-.0264743	.0564755
	_cons	-1.051382	.0224028	.001666	-1.051136	-1.094269	-1.008493
user_id	U0:sigma2	.003013	.0009442	.000204	.0028651	.0015461	.0054884
	U1:sigma2	.089927	.008486	.000889	.0894229	.074895	.1084871

Note: Default priors are used for model parameters.
Note: There is a high autocorrelation after 500 lags.
Note: Adaptation tolerance is not met in at least one of the blocks.

```
57 . timer off 5
58 . timer list 5
5: 753.24 / 1 = 753.2410
59 .
60 .
61 . *Total Timer off:
62 . timer off 9
63 . timer list 9
9: 2013.79 / 1 = 2013.7900
64 .
```